

SQL Developer Data Modeler



Oracle SQL Developer Data Modeler 4.1 - User defined DDL generation using transformation scripts

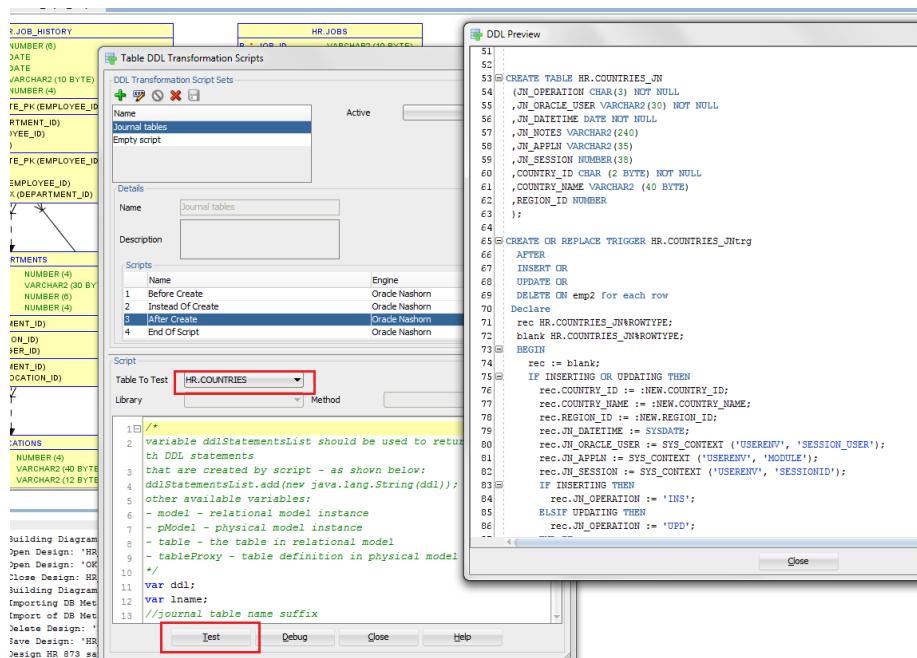
Gepostet von Philip Stoyanov-Oracle in SQL Developer Data Modeler am 12.03.2015 11:45:00

Data Modeler (DM) 4.0.3 (and all earlier versions) allows “Before Create”, “After Create” and “End of Scripts” scripts to be defined at table level and they will be included in generated DDL if related option is set. DM 4.0.3 introduced support for some variables to be used thus allowing building of templates - {table}, {table abbr}, {pk name}, {view}, {schema}, {long name}. Still there is a drawback that those scripts need to be added to tables.

DM 4.1 approached the problem in different way - was added support transformation scripts to be used at specific points of DDL generation for table - “Before Create”, “Instead of Create”, “After Create” and “End of Script”.

If “Instead of Create” script is defined it'll suppress built-in DDL generation for table and its components so DDL generation for non supported databases can be implemented.

Sets of such quadruples of scripts can be defined and one of them could be set as active – if no one is active then there is no custom DDL generation.



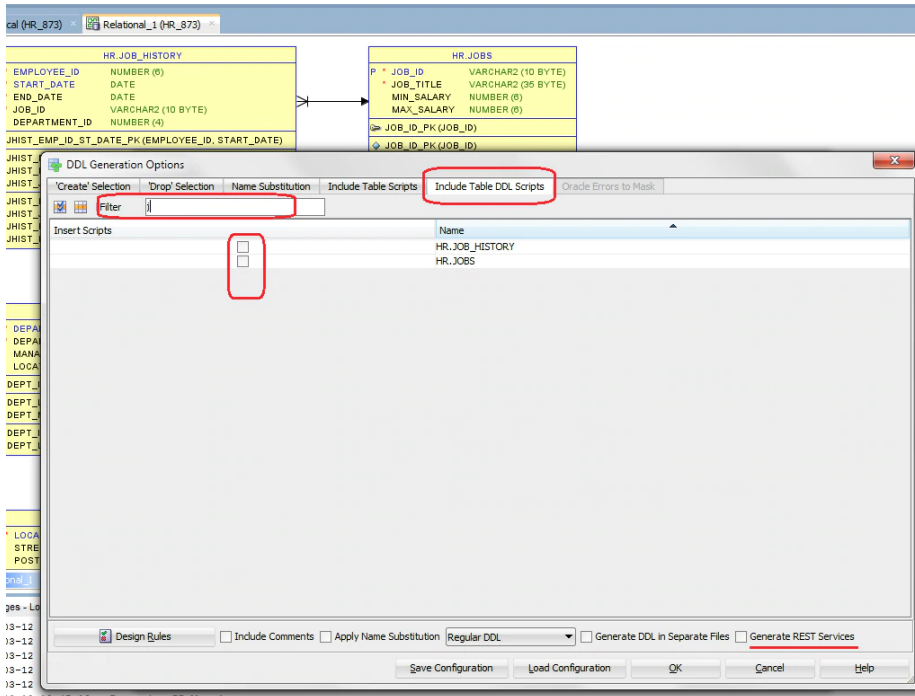
The dialog is activated from “Tools>Design rules and Transformations>Table DDL Transformations”. It allows writing, testing and debugging scripts. Table can be selected for “Test” and DDL preview window will pop-up and shows resulting DDL. There is no DDL preview in “Debug” mode.

Following variables are available to script during execution:

- **model** - relational model instance;
- **pModel** - physical model instance;
- **table** - the table in relational model;
- **tableProxy** - table definition in physical model;
- **ddlStatementsList** - should be used to return the list with DDL statements that are created by script – each DDL statement should be added as is shown below:

```
01. ddlStatementsList.add(new java.lang.String(ddl));
```

There is a new property on table defined “Use DDL Transformation Script” and if it's set then table will be included in custom DDL generation. That property can be set in table properties dialog or in “DDL generation options”



DM 4.1 is distributed with "After create" script that generates journal table and related supporting trigger providing 2 DDL statements for each processed table:

The script:

```

01.  /*
02.  variable ddlStatementsList should be used to return the list with DDL statements
03.  that are created by script - as shown below:
04.  ddlStatementsList.add(new java.lang.String(ddl));
05.  other available variables:
06.  - model - relational model instance
07.  - pModel - physical model instance
08.  - table - the table in relational model
09.  - tableProxy - table definition in physical model
10.  */
11.  var ddl;
12.  var lname;
13.  //journal table name suffix
14.  jnTabSuf = "_JN";
15.  // trigger name suffix
16.  jnTrgSuf = "_JNtrg";
17.  prompt = model.getAppView().getSettings().isIncludePromptInDDL();
18.  useSchema = model.getAppView().getSettings().isIncludeSchemaInDDL();
19.  if(model.getStorageDesign().isOpen()){
20.      if(useSchema){
21.          lname = tableProxy.getLongName();
22.      }else{
23.          lname = tableProxy.getName();
24.      }
25.  }else{
26.      if(useSchema){
27.          lname = table.getLongName();
28.      }else{
29.          lname = table.getName();
30.      }
31.  }
32.  if(prompt){
33.      ddl = "PROMPT Creating Journal Table for '"+lname+"';\n";
34.  }else{
35.      ddl = "";
36.  }
37.  ddl = ddl + "CREATE TABLE "+lname+jnTabSuf+"\n"+
38.  " (JN_OPERATION CHAR(3) NOT NULL"+"\n"+
39.  " ,JN_ORACLE_USER VARCHAR2(30) NOT NULL"+"\n"+
40.  " ,JN_DATETIME DATE NOT NULL"+"\n"+
41.  " ,JN_NOTES VARCHAR2(240)"+"\n"+
42.  " ,JN_APPLN VARCHAR2(35)"+"\n"+
43.  " ,JN_SESSION NUMBER(38)"+"\n";
44.  cols = table.getElements();
45.  for(var i=0;i<cols.length;i++){
46.      col = cols[i];
47.      ddl = ddl + " ,"+col.getName()+" "+col.getDatatypeString();
48.      if(col.isMandatory()){
49.          ddl = ddl + " NOT NULL\n";
50.      }else{
51.          ddl = ddl + "\n";
52.      }

```

```

53. }
54. ddl = ddl + " );";
55. ddlStatementsList.add(new java.lang.String(ddl));
56. if(prompt){
57.     ddl= "\n\nPROMPT Creating Journal Trigger for '"+lname+"';\n";
58. }else{
59.     ddl = "\n";
60. }
61. ddl = ddl +
62. "CREATE OR REPLACE TRIGGER "+lname+jnTrgSuf +" \n"+
63. " AFTER \n" +
64. " INSERT OR \n" +
65. " UPDATE OR \n" +
66. " DELETE ON emp2 for each row \n" +
67. " Declare \n" +
68. " rec "+lname+"_JN"+"%ROWTYPE; \n" +
69. " blank "+lname+"_JN"+"%ROWTYPE; \n" +
70. " BEGIN \n" +
71. "     rec := blank; \n" +
72. "     IF INSERTING OR UPDATING THEN \n" ;
73. for(var i=0;i<cols.length;i++){
74.     col = cols[i];
75.     ddl = ddl + "         rec."+col.getName()+" := :NEW."+col.getName()+" ; \n";
76. }
77. ddl = ddl +
78. "         rec.JN_DATETIME := SYSDATE; \n" +
79. "         rec.JN_ORACLE_USER := SYS_CONTEXT ('USERENV', 'SESSION_USER'); \n" +
80. "         rec.JN_APPLN := SYS_CONTEXT ('USERENV', 'MODULE'); \n" +
81. "         rec.JN_SESSION := SYS_CONTEXT ('USERENV', 'SESSIONID'); \n" +
82. "         IF INSERTING THEN \n" +
83. "             rec.JN_OPERATION := 'INS'; \n" +
84. "         ELSIF UPDATING THEN \n" +
85. "             rec.JN_OPERATION := 'UPD'; \n" +
86. "         END IF; \n" +
87. "         ELSIF DELETING THEN \n" ;
88. for(var i=0;i<cols.length;i++){
89.     col = cols[i];
90.     ddl = ddl + "         rec."+col.getName()+" := :OLD."+col.getName()+" ; \n";
91. }
92. ddl = ddl +
93. "         rec.JN_DATETIME := SYSDATE; \n" +
94. "         rec.JN_ORACLE_USER := SYS_CONTEXT ('USERENV', 'SESSION_USER'); \n" +
95. "         rec.JN_APPLN := SYS_CONTEXT ('USERENV', 'MODULE'); \n" +
96. "         rec.JN_SESSION := SYS_CONTEXT ('USERENV', 'SESSIONID'); \n" +
97. "         rec.JN_OPERATION := 'DEL'; \n" +
98. "     END IF; \n" +
99. "     INSERT into "+lname+"_JN" VALUES rec; \n" +
100. " END; \n" +
101. " /"
102. ddlStatementsList.add(new java.lang.String(ddl));

```

The transformation script also shows the usage of some settings in preferences - "Include Prompt" and 'Include Schema"

And here is the script generated In "DDL Preview" for table COUNTRIES:

```

01. CREATE TABLE HR.COUNTRIES
02. (
03.     COUNTRY_ID CHAR (2 BYTE)
04.     CONSTRAINT COUNTRY_ID_NN NOT NULL ,
05.     COUNTRY_NAME VARCHAR2 (40 BYTE) ,
06.     REGION_ID NUMBER ,
07.     CONSTRAINT COUNTRY_C_ID_PK PRIMARY KEY ( COUNTRY_ID )
08. )
09. ORGANIZATION INDEX
10. PCTTHRESHOLD 50
11. ;
12. /* ... other statements from generation ...*/
13. CREATE TABLE HR.COUNTRIES_JN
14. (JN_OPERATION CHAR(3) NOT NULL
15. ,JN_ORACLE_USER VARCHAR2(30) NOT NULL
16. ,JN_DATETIME DATE NOT NULL
17. ,JN_NOTES VARCHAR2(240)
18. ,JN_APPLN VARCHAR2(35)
19. ,JN_SESSION NUMBER(38)
20. ,COUNTRY_ID CHAR (2 BYTE) NOT NULL
21. ,COUNTRY_NAME VARCHAR2 (40 BYTE)
22. ,REGION_ID NUMBER
23. );
24.
25. CREATE OR REPLACE TRIGGER HR.COUNTRIES_JNtrg
26. AFTER
27. INSERT OR
28. UPDATE OR
29. DELETE ON emp2 for each row
30. Declare
31. rec HR.COUNTRIES_JN%ROWTYPE;
32. blank HR.COUNTRIES_JN%ROWTYPE;
33. BEGIN

```

```
34.      rec := blank;
35.      IF INSERTING OR UPDATING THEN
36.          rec.COUNTRY_ID := :NEW.COUNTRY_ID;
37.          rec.COUNTRY_NAME := :NEW.COUNTRY_NAME;
38.          rec.REGION_ID := :NEW.REGION_ID;
39.          rec.JN_DATETIME := SYSDATE;
40.          rec.JN_ORACLE_USER := SYS_CONTEXT ('USERENV', 'SESSION_USER');
41.          rec.JN_APPLN := SYS_CONTEXT ('USERENV', 'MODULE');
42.          rec.JN_SESSION := SYS_CONTEXT ('USERENV', 'SESSIONID');
43.          IF INSERTING THEN
44.              rec.JN_OPERATION := 'INS';
45.          ELSIF UPDATING THEN
46.              rec.JN_OPERATION := 'UPD';
47.          END IF;
48.          ELSIF DELETING THEN
49.              rec.COUNTRY_ID := :OLD.COUNTRY_ID;
50.              rec.COUNTRY_NAME := :OLD.COUNTRY_NAME;
51.              rec.REGION_ID := :OLD.REGION_ID;
52.              rec.JN_DATETIME := SYSDATE;
53.              rec.JN_ORACLE_USER := SYS_CONTEXT ('USERENV', 'SESSION_USER');
54.              rec.JN_APPLN := SYS_CONTEXT ('USERENV', 'MODULE');
55.              rec.JN_SESSION := SYS_CONTEXT ('USERENV', 'SESSIONID');
56.              rec.JN_OPERATION := 'DEL';
57.          END IF;
58.          INSERT into HR.COUNTRIES_JN VALUES rec;
59.      END;
60.  /
```

Philip Stoyanov

5992 Ansichten

Tags:

Durchschnittliche Benutzbewertung

(0 Bewertungen)

1 Kommentar

**Philip Stoyanov-Oracle** 03.11.2015 17:19

there is an error in line 66 of the JavaScript :
the wrong line 66:

" DELETE ON emp2 for each row \n" +

correct line 66:

" DELETE ON "+lname+" for each row \n" +

Philip

Aktionen

Gefällt mir (1)

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